Alireza Karami

Undergraduate Researcher in Artificial Intelligence and Data Science for Healthcare

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RESEARCH PROFILE

Computer Science undergraduate at Taylor's University, specialising in Artificial Intelligence with an extension in Data Science, working on machine learning for healthcare. First author of two papers and co-first author of one: one accepted at ICCRD 2027 and two under peer review. Experienced in explainable clinical prediction models, privacy-preserving federated learning for medical imaging and healthcare analytics dashboards, with hands-on laboratory work on extended-reality headsets.

Research interests: explainable AI for clinical decision support; federated and privacy-preserving machine learning; medical image analysis; healthcare operations analytics.

EDUCATION

Bachelor of Computer Science (Artificial Intelligence), Extension in Data Science Expected Feb 2028

Taylor's University, Selangor, Malaysia

CGPA: 3.30 / 4.00

Relevant coursework: Data Mining, Data Science Principles, Big Data Technologies, Computer Vision and Natural Language Processing, Computer Cognitive Applications, Advanced Programming, Data Structures and Algorithms, Database Systems

PUBLICATIONS

ACCEPTED

[1] An Intelligent Healthcare Analytics Framework for Emergency Department Optimization Using AI and Visual Dashboards

A. Karami*, et al. In *Proceedings of the 19th International Conference on Computer Research and Development (ICCRD 2027)*, Singapore, 15–17 January 2027. Accepted for presentation and publication.

UNDER REVIEW

[2] MediPro: A Privacy-Preserving AI-Based Federated Learning System for Neurosurgical Disease Prediction and Clinical Decision Support

A. Karami (first author), et al. Manuscript under peer review, 2026.

[3] Multimodal Artificial Intelligence for Personalized Seizure Risk Stratification and Clinical Decision Support in Traumatic Brain Injury

A. Karami (first author), et al. Submitted to the *3rd International Conference on AI in Medicine and Healthcare (AiMH 2027)*, Malé, Maldives. Under review.

** Equal contribution (co-first author). All papers supervised by Dr. Soobia Saeed.*

RESEARCH EXPERIENCE

Undergraduate Researcher, AI for Healthcare 2026 – Present

Research Supervisor: Dr. Soobia Saeed

- **Federated learning for medical imaging [2], first author.** Proposed MediPro, a clinical decision support system that trains a shared CNN across institutions without exchanging patient data. Fine-tuned DenseNet121 on public brain-tumour, bone-fracture and skin-disease image datasets split across simulated clients and aggregated with Federated Averaging (FedAvg), reaching **~75–80% accuracy** with a **2–3 s** prediction time.
- **Secure system design [2].** Designed the MediPro architecture (Django back end, MongoDB database, mobile front end) with AES-256 and TLS encryption, role-based access control and audit logging.
- **Seizure-risk prediction in traumatic brain injury [3], first author.** Co-developed an AI clinical decision support system (AI-CDSS); compared Logistic Regression, Random Forest, SVM and XGBoost under stratified five-fold cross-validation, with XGBoost performing best (**~92% accuracy, AUC 0.95**); applied SHAP to explain predictions for personalised risk stratification.

- **Emergency-department analytics [1], co-first author.** Co-developed an AI-based analytics framework with interactive dashboards for patient-flow monitoring and resource allocation; contributed to the research design, literature review, data analysis, dashboard development and manuscript preparation.

LABORATORY EXPERIENCE

Part-Time Lab Assistant Sep 2025 – Present

Vortex Lab, Taylor's University, Malaysia

- Work hands-on with extended-reality (XR) headsets, including **Meta Quest** and **Microsoft HoloLens**, setting up and testing the devices for lab projects.
- Set up, maintain and organise laboratory equipment and computing resources, and pre-process data for ongoing lab activities.
- Support students during lab sessions by troubleshooting programming and technical issues.
- Collaborate with lab members and supervisors on documentation and shared technical work.

SELECTED PROJECTS

Flood Management System Linux, Cloudera (Hadoop)

- Personal project: reviewed approaches to flood-risk control and management, then built a data-processing workflow on the Cloudera Hadoop platform using Linux command-line tools.

Air Pollution Control System Java

- Course project: built a Java application to monitor air-pollution levels and to manage and analyse air-quality data.

Website Prototype Web development

- Course project: planned the layout and user interface, then built a working prototype of the core features using object-oriented programming.

TECHNICAL SKILLS

Programming	Python, Java, JavaScript, R
Machine learning	CNNs, transfer learning (DenseNet121), federated learning (FedAvg), XGBoost, Random Forest, SVM, Logistic Regression
Research methods	Stratified k-fold cross-validation, SHAP explainability, literature review, manuscript preparation
Data & systems	Data analysis, data mining, visual dashboards, Cloudera (Hadoop), Google Colab, Linux command line
Web development	HTML, CSS, Next.js, Django, MongoDB
Extended reality	Meta Quest, Microsoft HoloLens

HONOURS, ATHLETICS & LEADERSHIP

National Badminton Athlete, Iran 2012 – 2024

- Member of Iran's national badminton team for seven years, competing in international events including the Fajr International Badminton Tournament.
- Competed at an Asian youth badminton championship in Indonesia (2016) and at the Badminton Asia Championships in Wuhan, China (2017).
- Represented the national team at an international competition in Baku, Azerbaijan (2018).

Silver Medal, Men's Doubles: Taylor's University Badminton Competition Jul 2025

Badminton Coach

- Coached players for several years, supporting skill development, training discipline and competitive readiness.

LANGUAGES & CERTIFICATIONS

Languages	Persian (native); English (professional working proficiency, IELTS Academic 6.5, 2024); Japanese (beginner)
Certification	Introduction to Agents (Mercor)